

4 (a) (i)

$$\sin \theta = \frac{\lambda}{b}$$

$$\sin(0.16^\circ) = \frac{\lambda}{0.24 \times 10^{-3}}$$

$$\lambda = 6.70 \times 10^{-7} \text{ m [B1]}$$

(ii) Distance from centre to first minimum, y :

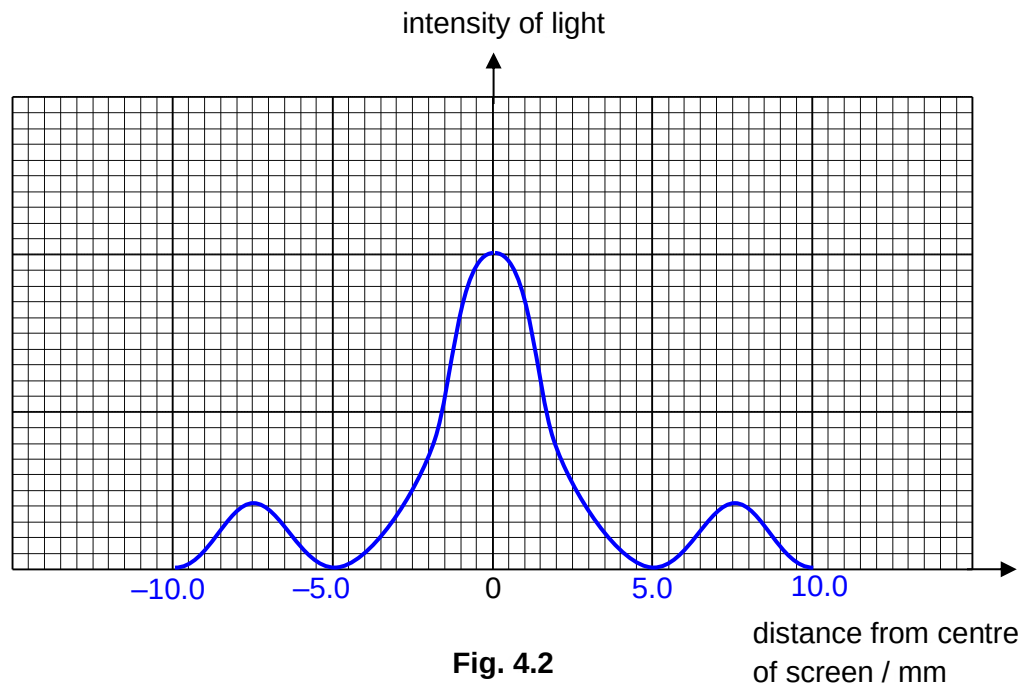
$$\tan \theta = \frac{y}{D}$$

$$y = (1.8) \tan(0.16^\circ) = 5.0 \times 10^{-3} \text{ m}$$

$$= 5.0 \text{ mm [M1]}$$

Width of central maximum = $2y = 10.0 \text{ mm [A1]}$

(iii)



[B1] correct shape

[B1] correct labels on horizontal axis + distance between minima = distance from centre to 1st minimum

(iv) The minimum will be further away from the centre / The central maximum will spread out more. [B1]
The intensity will be reduced. [B1]

(b) (i) Light waves from the sources have a constant phase difference (and same frequency) [B1]

(ii) Fringe separation, x :

$$x = \frac{\lambda D}{a} = \frac{(638 \times 10^{-9})(1.8)}{0.82 \times 10^{-3}} \text{ [M1]}$$

$$= 1.40 \times 10^{-3} \text{ m} = 1.40 \text{ mm [M1]}$$

distance to 2nd dark fringe = $1.5x = 2.10 \text{ mm [A1]}$

(iii) Resultant amplitude at bright fringe = $A + A = 2A$ [C1]

$$I \propto A^2$$

$$\frac{I_{\text{fringe}}}{I} = \frac{A_{\text{fringe}}^2}{A^2} = \frac{(2A)^2}{A^2} \quad [\text{M1 - awarded for concept of proportionality}]$$

$$I_{\text{fringe}} = 4I \quad [\text{A0}]$$

(iv) Although intensity of bright fringe is $4I$, intensity at dark fringes are zero. [B1]

Hence average intensity of the interference pattern is $2I$. [B1]

Principle of conservation of energy is not violated.